MTH219 2020 S2 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ICT133 Structured Programming
* ICT162 Structured Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

Question 1

(a) It was found that when children tell stories, they tend to recall stories as a sequence of actions rather than the overall plot. Their descriptions of the stories are filled with the phrase "and then …". Suppose an experimenter asked a random sample of 15 students to tell her about a particular movie they have watched. Among other variables, she counted the number of "and then …" statements. The data is shown in Table Q1(a).

22	23	19	21	20
18	24	43	27	28
20	23	26	23	19

Table Q1(a)

(i) Determine the following sample statistics:

Mean, Median, Mode, Standard deviation, Variance, Range, First quartile, Third quartile, Inter-quartile range, Coefficient of Variation.

(14 marks)

(ii) If the skewness statistic is found to be 2.44, comment on the data distribution.

(1 mark)

(iii) Calculate the 80th percentile.

(2 marks)

(b) Describe an example of a small set of data for which the mean is larger than the third quartile.

(3 marks)

1(a) i.

$$X = \left\{ \begin{array}{lllll} 18, & 19, & 19, & 20, & 20, \\ 21, & 22, & 23, & 23, & 23, \\ 24, & 26, & 27, & 28, & 43, \end{array} \right\}$$

min X = 18

max X = 43

range of X = max X − min X = 25

First Quartile = $X_{0.25(15+1)}$

= X_4

= 20

Third Quartile = $X_{0.75(15+1)}$

= X_{12}

= 26

Inter Quartile Range = 3rd Quartile − 1st Quartile

= $X_{12} - X_4$

= 6

1(a) i. median $= X_{0.50(15+1)}$

$= X_8$

$= 23$

mode $= 23$

mean $= \bar{x}$

$= \frac{1}{n}\sum x$

$= \frac{1}{15}(356)$

$= 23.73$

variance $= s^2$

$= \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$

$= \frac{1}{14}(522.9333)$

$= 37.35$

standard deviation $= s$

$= \sqrt{37.35}$

$= 6.11$

coeff of variation $= \frac{s}{\bar{x}}$

$= \frac{6.11}{23.73}$

$= 0.2575$

ii. skewness $= 2.44$

A positive skewness indicates more data points are above the mean than data points below the mean.

The long tail of the distribution is on the right.

iii 80th percentile $= X_{0.80(15+1)}$

$= X_{12.8}$

$= X_{12} + 0.8 \times (X_{13} - X_{12})$

$= 26 + 0.8(27-26)$

$= 26.8$

(b) mean > 3rd Quartile

consider sample size 7

3rd Quartile $= X_{0.75(7+1)}$

$= X_6$

$\Rightarrow$ assign low values for X_1 to X_5

$\Rightarrow$ assign high value for X_7

$\therefore$ data set $= \{0, 0, 0, 1, 1, 1, 11\}$

$\bar{x} = 2$

$X_6 = 1$

$\bar{x} > X_6$

Question 2

(a) Suppose we roll one unbiased six-sided die over and over again. Compute the number of rolls needed so that the probability of obtaining the number "4" within this number of rolls is at least 50 %. Explain the probability model.

(6 marks)

(b) In a tennis match, the player who is serving has two chances to hit the ball towards the opponent's end. According to the rules, the ball must fall in the correct marked box area on the opposite side of the net. A serve that misses this marked box will be considered a "fault". In the event that the player misses both serves, then this will be considered as a "double fault" and as a result, the player will lose a point. Suppose during the entire tennis tournament, based on the statistics, the champion Mr Tan successfully served 65 % of his first serves. Given that he made a "fault" with his first serve, he also made a "fault" on his second serve 8 % of the time. Compute the probability that Mr Tan makes a "double fault" when he serves.

(3 marks)

(c) Mr Lee rolled two dice. Suppose he knows that one of the dice turned up a face value number "3". Compute the probability that the other die also turned up the number "3".

(4 marks)

(d) Suppose a particular mother's behaviour is independent of her baby. The mother looks at her baby a total of 4 hours each day, and the baby looks at the mother a total of 3 hours each day. Assume that both the mother and child are asleep from 10pm to 8am. Compute the probability that both the mother and baby will look at each other at the same time.

(3 marks)

(e) A study was conducted on 4000 Dalmatians examined for both hearing impairment and whether iris colour is blue. It was found that 30 % of the Dalmatians were hearing impaired, 12 % were blue-eyed, 6 % were hearing impaired and blue eyed. If a Dalmatian is randomly selected, compute the probability that it is either hearing impaired or blue eyed.

(2 marks)

(f) Describe the following variables as discrete or continuous.

(i) Waiting time for bus service number "154".

(1 mark)

(ii) Number of complaints received.

(1 mark)

2(a) $P(\text{roll} = 4) = \frac{1}{6}$ $\quad P(\text{roll} \neq 4) = \frac{5}{6}$

Since there are only 2 outcomes of concern, we apply the binomial model

let Y be the number of rolls of 4

$$Y \sim Bin(n, \tfrac{1}{6})$$

$$P(Y \geq 1) \geq 0.50$$
$$1 - P(Y=0) \geq 0.50$$
$$P(Y=0) \leq 0.50$$
$$\left(\tfrac{5}{6}\right)^n \leq 0.50$$
$$n \ln\left(\tfrac{5}{6}\right) \geq \ln 0.50$$
$$n \geq \frac{\ln 0.50}{\ln\left(\frac{5}{6}\right)}$$
$$n \geq 3.80$$

$\therefore$ The number of rolls needed is 4

(b)

$P(\text{1st serve success}) = 0.65$

$P(\text{1st serve fault}) = 0.35$

$P(\text{2nd serve fault} \mid \text{1st serve fault}) = 0.08$

$$\frac{P(\text{1st serve fault} \cap \text{2nd serve fault})}{P(\text{1st serve fault})} = 0.08$$

$$P(\text{1st serve fault} \cap \text{2nd serve fault}) = 0.08 \times P(\text{first serve fault})$$
$$= 0.08 \times 0.35$$
$$= 0.028$$

$\therefore P(\text{double fault}) = 0.028$

2(c) one of the dice is 3 = $\{(3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (1,3), (2,3), (4,3), (5,3), (6,3)\}$

both dices are 3 = $\{(3,3)\}$

$P(\text{the other dice is 3} \mid \text{one of the dice is 3})$

$= \dfrac{P(\text{both dices are 3})}{P(\text{one of the dice is 3})}$

$= \dfrac{1}{11}$

2(d) length of day = 14h

$P(\text{mother looks at baby}) = \dfrac{4}{14} = \dfrac{2}{7}$

$P(\text{baby looks at mother}) = \dfrac{3}{14} = \dfrac{3}{14}$

P(Mother and baby look at each other at the same time)

= P(Mother looks at baby) × P(Baby looks at mother)

$= \dfrac{2}{7} \times \dfrac{3}{14}$

$= \dfrac{3}{49}$

2(e) D: the dalmantian is deaf

B: the dalmantian has blue iris

$P(D) = 0.3$

$P(B) = 0.12$

$P(D \cap B) = 0.06$

$P(D \cup B) = P(D) + P(B) - P(B \cap D)$

$= 0.3 + 0.12 - 0.06$

$= 0.36$

2(f) i. continuous

ii. discrete

Question 3

(a) Suppose a volume of 1000 drops of water contains 2000 bacteria, separate from each other and thoroughly mixed in the water. A single drop of water is now smeared uniformly over the surface of a dish. After 7 days, wherever a bacterium was deposited on the dish, a visible colony of bacteria appears. Assume that the positions of bacteria colonies over the plate can be adequately modelled by a Poisson distribution.

(i) Apply the Poisson model and compute the probability that one bacteria colony appears on the whole plate.

(3 marks)

(ii) Compute the probability that the number of bacteria colonies that appear on the whole plate is more than 2.

(3 marks)

(iii) Compute the probability two bacteria colonies appears over an area of half the plate.

(3 marks)

(b) The probability of the number of customers (denoted as X) who will be put on hold when they call the service centre hotline is shown in Table Q3(b) as P(X). Calculate the mean and standard deviation of X.

(6 marks)

X	1	4	2	0	3
P(X)	0.34	0.04	0.23	0.18	0.21

Table Q3(b)

(c) Suppose the diameter of a type of construction cables is normally distributed with a mean of 5.8 mm and a standard deviation of 0.07 mm. A cable is considered unusable if its diameter is smaller than 5.65 mm or larger than 5.98 mm. On the other hand, if the cable diameter is between 5.94 mm and 5.98 mm, it is certified as usable.

(i) Compute the probability that a randomly selected cable is usable.

(2 marks)

(ii) Compute the probability that a randomly selected cable is unusable.

(3 marks)

3(a) i. $\lambda = \frac{2000}{1000} = 2$

let Y be the no. of bacteria in a drop

$Y \sim Po(2)$

P(1 bacteria along)

$= P(Y=1)$

$= \frac{e^{-2} \cdot 2^1}{1}$

$= 0.27$

ii. $P(Y=0) = \frac{e^{-2} \cdot 2^0}{0!} = 0.135$

P(>2 colonies) = 1 - P(≤2 bacteria colonies)

$= 1 - P(Y \leq 2)$

$= 1 - \text{ppois}(2, 2)$

$= 0.32$

iii. let X be no. of bacteria in $\frac{1}{2}$ a plate

$\lambda_X = \frac{1}{2}(\lambda) = 1$

$X \sim Po(1)$

P(2 bacteria colonies appear on half a plate)

$= P(X=2)$

$= \frac{e^{-1} \lambda_x^2}{2!}$

$= 0.18$

3(b) mean $= \sum_{k=0}^{4} k P(X=k)$

$= 1.59$

variance $= \sum_{k=0}^{4} (k-1.59)^2 P(X=k)$

$= 1.2619$

standard deviation $= \sqrt{\text{variance}}$

$= \sqrt{1.2619}$

$= 1.12$

3(c) let D be the diameter of the cable

$D \sim N(5.8, 0.07^2)$

i. $P(\text{cable usable}) = P(5.94 \leq D \leq 5.98)$

$= P(D \leq 5.98) - P(D \leq 5.94)$

$= \text{pnorm}(5.98, 5.8, 0.07) - \text{pnorm}(5.94, 5.8, 0.07)$

$= 0.0177$

ii. $P(D > 5.98) = 1 - P(D \leq 5.98)$

$= 1 - \text{pnorm}(5.98, 5.8, 0.07)$

$= 0.005063995$

$P(D < 5.65) = P(D \leq 5.65)$

$= \text{pnorm}(5.65, 5.8, 0.07)$

$= 0.01601229$

$P(\text{cable is unusable}) = P(D > 5.98 \text{ or } D < 5.65)$

$= P(D > 5.98) + P(D < 5.65)$

$= 0.0211$

Question 4

(a) Suppose two unbiased six-sided dice are thrown. Let X denotes the event that the number shown by the first die is even. Let Y denotes the event that the sum of the dice is odd. Show that both X and Y are independent events.

(4 marks)

(b) Suppose in order to qualify for the officer cadet course, candidates must score in the top 15 % on a special academic test. The test score has a mean of 200 and a standard deviation of 22. Assume that the test scores can be modelled by a normal distribution. Determine the lowest possible score to qualify.

(4 marks)

(c) Given that the amount of time that an employee of company XYZ stays in a demanding job can be modelled by exponential distribution with a mean of 26 months.

(i) Compute the probability that the employee has to be replaced within 7 months.

(2 marks)

(ii) Compute the probability that the employee has to be replaced during the third year.

(3 marks)

(d) Consider the following pdf of X:

$$f(x) = mx^3 \qquad \text{where } 0 \leq x \leq 3$$

Solve for the unknown m.

(3 marks)

(e) Suppose an electrical signal's trajectory X can be adequately modelled by a continuous uniform distribution on the interval (10, 25).

(i) Describe the pdf and cdf.

(2 marks)

(ii) Compute the probability that the trajectory falls between 14 and 20.

(2 marks)

4(a) let D_1 be the dice roll of the first dice

let D_2 be the dice roll of the 2nd dice

$$D_1 = \{1,2,3,4,5,6\}, \quad D_2 = \{1,2,3,4,5,6\}$$

$$P(X) = P(D_1 \text{ is even})$$
$$= \frac{3}{6}$$
$$= \frac{1}{2}$$

$$P(Y) = P(D_1 + D_2 \text{ is odd})$$
$$= P((D_1 \text{ even and } D_2 \text{ odd}) \text{ or } (D_1 \text{ even and } D_2 \text{ odd}))$$
$$= P(D_1 \text{ even and } D_2 \text{ odd}) + P(D_1 \text{ even and } D_2 \text{ odd})$$
$$= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}$$
$$= \frac{1}{2}$$

$$P(D_1 \text{ even and } D_2 \text{ odd}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$
$$P(X)P(Y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Hence, events X and Y are independent

(b) let Y be the test score

$Y \sim N(200, 22^2)$

min. score = 85th percentile

= qnorm(0.85, 200, 22)

= 222.8

4(c) let Y be the time employee stays with the job

P(employee replaced within 7 months)

$= P(X \leq 7)$

$= \text{pexp}(7, \frac{1}{26})$

$= 0.236$

P(employee replaced within 3rd year)

$= P(X \leq 36) - P(X \leq 24)$

$= \text{pexp}(36, \frac{1}{26}) - \text{pexp}(24, \frac{1}{26})$

$= 0.14687$

4(d) $f(x) = mx^3$ for $0 \le x \le 3$

$$\int_0^3 f(x)\,dx = 1$$

$$\int_0^3 mx^3\,dx = 1$$

$$\frac{m}{4}\left[x^4\right]_0^3 = 1$$

$$\therefore m = \frac{4}{\left[x^4\right]_0^3} = \frac{4}{81}$$

(e) $X \sim$ uniform $[10, 25]$

i. $\text{pdf} = \frac{1}{25-10} = \frac{1}{15}$

$\text{cdf} = \frac{x-10}{25-10} = \frac{x-10}{15}$

ii. $P(14 \le X \le 20)$

$$= \frac{20-14}{25-10}$$

$$= \frac{6}{15}$$

$$= \frac{2}{5}$$

Question 5

(a) Assume that the probability that a coin lands on its edge is 0.001. Suppose you flip a coin 3000 times.

(i) Apply a suitable probability model and compute the exact probability that 5 flips will land on their edge.

(3 marks)

(ii) By applying the Poisson model as an approximation, solve the probability that 5 flips will land on their edge. Comment on the results.

(5 marks)

(b) A researcher wishes to select people in the middle 50 % of the population based on blood pressure for a medical study. Suppose the mean systolic blood pressure is 120 and the standard deviation is 9, determine the upper and lower readings that would qualify people to participate.

(4 marks)

(c) A manufacturer of washing machine purchases a particular microcontroller from 3 suppliers: Company A, Company B and Company C. Given that 30 % of these microcontrollers are purchased from Company A, 20 % from Company B and the remaining 50 % from Company C. From past historical records, 5 % of the microcontrollers from Company B are defective, 4 % of the microcontrollers from Company C are defective, and 3 % of the microcontrollers from Company A are defective. When the microcontrollers arrive from the 3 suppliers, they are placed directly in a bin and not inspected. Suppose an engineer randomly selects a microcontroller for testing and discovers that it is defective. Compute the probability that this microcontroller was manufactured by Company B.

(8 marks)

5(a) $p = 0.001$

i. let X be the number of flips where the coin landed on the edge in 3000 flips

$$X \sim \text{Binomial}(3000, 0.001)$$

P(5 flips will land on the edge)

$= P(X = 5)$

$= \binom{3000}{5} p^5 (1-p)^{2995}$

$= \text{dbinom}(5, 3000, 0.001)$

$= 0.100856$

$\simeq 0.101$

ii. let Y be the binomial approximation for X

$np = 3000 \times 0.001 = 3$

$Y \sim Po(3)$

$P(X=5) \simeq P(Y=5)$

$= \text{dpois}(5, 3)$

$= 0.101$

The approximation is valid because

1) events occur singly
2) the average rate of occurance remains constant
3) events happen independently of each other

5(b) let X be the systolic blood pressure

$X \sim N(120, 9^2)$

lower reading = 25th percentile

= qnorm(0.25, 120, 9)

= 113.93 mmHg

upper reading = 75th percentile

= qnorm(0.75, 120, 9)

= 126.07 mmHg

5(c) A: Microcontroller is manufactured by supplier A

B: Microcontroller is manufactured by supplier B

C: Microcontroller is manufactured by supplier C

D: Microcontroller is defective

$P(A) = 0.3$ $P(B) = 0.2$ $P(C) = 0.5$

$P(D|A) = 0.03$ $P(D|B) = 0.05$ $P(D|C) = 0.04$

$P(A \cap D) = P(A) \cdot P(D|A) = 0.0009$

$P(B \cap D) = P(B) \cdot P(D|B) = 0.01$

$P(C \cap D) = P(C) \cdot P(D|C) = 0.002$

$P(D) = P(A \cap D) + P(B \cap D) + P(C \cap D) = 0.039$

$$P(B|D) = \frac{P(B \cap D)}{P(D)} = \frac{0.01}{0.039} = 0.256$$